

Metric	GSM8K			Math			OlympiadBench			Omni-MATH			Overall	
	TNR	TPR	F1	TNR	TPR	F1	TNR	TPR	F1	TNR	TPR	F1	F1	TFLOPs
<i>Qwen3-1.7B</i>														
Basic Prompt	17.9	96.4	30.2	14.5	84.2	24.7	7.7	78.8	14.1	8.0	82.6	14.7	20.9	2.86
<i>Probe-only</i>	22.2	80.3	34.8	17.8	66.5	28.1	13.2	52.5	21.0	10.4	61.0	17.8	25.4	3.05
Prompt Opt.	22.2	76.7	34.5	18.0	58.4	27.5	13.3	49.6	21.0	10.4	57.3	17.6	25.1	3.19
+Cons@4 [†]	21.6	81.2	34.1	20.3	69.4	31.4	11.0	58.5	18.5	12.0	62.2	20.0	26.0	12.71
VerifySteer-Uni	25.6	77.2	38.5	21.7	61.6	32.1	15.6	56.6	24.4	12.6	59.3	20.9	29.0	3.37
VerifySteer-Bi	27.1	83.4	40.9	21.5	69.2	32.9	15.0	58.4	23.8	12.6	64.3	21.1	29.7	3.47
<i>Qwen3-8B</i>														
Basic Prompt	54.6	96.4	69.7	58.4	83.0	68.6	54.0	76.1	63.2	53.8	73.4	62.1	65.9	16.71
<i>Probe-only</i>	57.0	94.3	71.1	62.0	87.9	72.7	56.4	79.9	66.2	57.6	76.8	65.8	69.0	18.71
Prompt Opt.	58.5	94.3	72.2	66.8	85.0	74.8	58.4	77.9	66.7	58.6	75.5	66.0	69.9	21.26
+Cons@4 [†]	63.3	95.6	76.2	69.5	86.3	77.0	61.3	77.9	68.7	63.1	71.4	67.0	72.2	84.09
VerifySteer-Uni	62.3	93.3	74.7	69.0	85.5	76.4	59.6	77.0	67.2	61.9	76.8	68.5	71.7	21.23
VerifySteer-Bi	64.3	93.8	76.3	68.7	86.9	76.7	60.1	76.7	67.4	61.8	77.2	68.6	72.3	21.23
<i>Qwen3-14B</i>	61.4	98.4	75.6	68.5	86.9	76.6	58.9	74.3	65.7	59.6	80.2	68.4	71.6	37.45
<i>Llama-3.3-70B</i>	68.6	96.9	80.3	46.0	94.3	61.8	32.8	93.2	48.6	29.0	88.0	43.6	58.6	85.45
<i>Qwen2.5-72B</i>	61.8	95.9	75.2	45.8	91.4	61.0	35.9	87.6	50.9	37.5	83.4	51.8	59.7	53.03

Table 2: Comparison of VerifySteer with baselines on ProcessBench. Cons@4 refers to self-consistency method with 4 sampled results and the final answer is selected by majority voting. [†] denotes sampling-based decoding with temperature 0.7 and top- p 0.8. Methods without [†] use greedy decoding. TFLOPs reports the estimated inference compute per sample.

- Updated Table 2 as supporting material for COLM rebuttal, new results are marked in red.